

MD. MAHMUDUL HASAN

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RESEARCH INTERESTS

Biomedical Signal Processing • Machine Learning & Deep Learning • Edge AI • VLSI & AI-Hardware Systems

EDUCATION

Chittagong University of Engineering & Technology (CUET)	2023 – Present
<i>B.Sc. in Electrical and Electronic Engineering</i>	<i>CGPA: 3.50</i>
	<i>Last Semester CGPA: 3.68</i>

Notre Dame College, Dhaka	2020 – 2022
<i>Higher Secondary Certificate (HSC)</i>	<i>GPA: 5.00/5.00 100% Attendance</i>

NKM High School and Homes	2010 – 2020
<i>Secondary School Certificate (SSC)</i>	<i>GPA: 5.00/5.00 General Grade Scholarship</i>

RESEARCH EXPERIENCE

SASA-Net: Lightweight DCT-Domain Spectral Attention for Inter-Patient ECG Arrhythmia Classification
Manuscript submitted

Illumination-Robust Image Processing and Cross-Domain Analysis for Conjunctival Anemia Assessment
Manuscript submitted

PROJECTS

ESP32-Based Smart Digital Thermometer with Real-Time Alert System
Integrated an ESP32 microcontroller, DS18B20 sensor, OLED display, and buzzer for real-time monitoring.

Water Level Alarming with Water Pump Control
Automated pump control circuitry using a 555 timer, water-level probes, and a relay.

305 kVA Distribution Transformer Design
Calculated core dimensions, winding specifications, and loss parameters for a distribution transformer.

Interactive Engineering Tools
Developed a multi-variable Karnaugh Map Solver in Python and interactive educational tools for digital modulation and JFET physics.

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, Assembly
Signal Processing: Digital Signal Processing, Biomedical Signal Processing
AI / ML: Machine Learning, Deep Learning
VLSI & Digital Design: VLSI, Digital Logic Design
Embedded: Embedded Systems, Arduino, ESP32
Engineering Tools: PSpice, Proteus, AWR, ADS

LEADERSHIP & ACTIVITIES

Vice Chair | IEEE Power and Energy Society, CUET Chapter

Robi Data Privacy Avengers 2026

SUROKKHA — Continuous Data & AI Trust Loop

IEEE CUET SB SciBlitz 2025 — Three-Minute Thesis (3MT)

Participant

IEEE CUET SB Face The Case 4.0

Participant